

This is a great project because it can showcase **Amazon Lex, AWS Lambda, DynamoDB, API integration, serverless architecture, and AI-powered learning.** Breaking it into phases will make it much easier to build and demonstrate on your CV or portfolio.

Project: AWS Cloud Practitioner Quiz Assistant

Phase 1: Foundation MVP

Objective

Build a basic chatbot that can ask AWS Cloud Practitioner quiz questions.

Features

- Amazon Lex chatbot
- Welcome message
- User selects:
 - Service Category (Compute, Storage, Database, Security, etc.)
 - Difficulty (Easy, Intermediate, Difficult)
 - Number of Questions
- Questions stored in DynamoDB
- Lambda retrieves questions
- Score tracking
- Final score displayed

AWS Services

- Amazon Lex
- AWS Lambda
- Amazon DynamoDB
- Amazon CloudWatch

Deliverable

A working chatbot capable of conducting a quiz and calculating a score.

Phase 2: Learning Mode

Objective

Transform the chatbot from a simple quiz engine into a learning platform.

Features

After each question:

✅ Explain why the correct answer is correct

❌ Explain why each incorrect answer is wrong

💡 Display exam tips

Example

Question:

Which AWS service provides virtual servers?

Response:

- Why EC2 is correct
- Why S3 is incorrect
- Why RDS is incorrect
- Why CloudFront is incorrect

Database Enhancement

Store:

- Correct answer explanation
- Incorrect answer explanations
- Exam tips

Deliverable

Users learn concepts instead of merely memorizing answers.

Phase 3: Advanced Topic Selection

Objective

Allow users to customize their study path.

Conversation Flow

Which category?

- Compute
- Storage
- Database
- Networking
- Security

If Compute:

Which service?

- EC2
- ECS
- EKS
- Lambda
- Fargate
- All Compute Services

Then:

- Difficulty
- Number of Questions

Deliverable

Personalized learning experience.

Phase 4: Full Mock Exam Engine

Objective

Simulate the actual AWS Cloud Practitioner exam.

Features

Exam Mode:

- No hints
- No explanations during the test
- Timer (optional)
- Question randomization

Final Report:

- Score
- Percentage
- Weak areas

Example

Score: 42/65

Areas to Improve:

- IAM
- VPC

- ECS

Deliverable

A realistic exam simulation.

Phase 5: Progress Tracking

Objective

Track user performance over time.

Features

Store:

- User name
- Quiz date
- Score
- Services studied
- Weak topics

Dashboard

Build a dashboard using:

- Amazon QuickSight

Metrics:

- Average score
- Most studied services
- Weakest services
- Improvement trend

Deliverable

A complete learning analytics platform.

Phase 6: AI Question Generation

Objective

Generate new questions automatically.

Features

Use:

- Amazon Bedrock

Prompt example:

Generate 10 AWS Cloud Practitioner questions about EC2 at Intermediate level.

AI returns:

- Question
- Options
- Correct answer
- Explanations
- Exam tips

Benefits

- Unlimited questions
- Reduced manual maintenance
- Keeps content fresh

Deliverable

An AI-powered quiz engine.

Phase 7: Certification Coach

Objective

Turn the chatbot into a study assistant.

Features

Ask:

- Exam date
- Target score
- Available study hours

Generate:

- Weekly study plan
- Recommended AWS services
- Practice quizzes

Example:

Week 1:

- Cloud Concepts
- EC2
- S3

Week 2:

- IAM
- VPC
- Billing

Deliverable

A personalized AWS certification coach.

Suggested Timeline

Phase	Duration
Phase 1 - MVP Quiz Bot	1 Week
Phase 2 - Explanations	1 Week
Phase 3 - Topic Selection	3 Days
Phase 4 - Mock Exam	1 Week
Phase 5 - Progress Tracking	1 Week
Phase 6 - Bedrock AI Integration	1-2 Weeks
Phase 7 - Certification Coach	1 Week

By the end, you'll have a portfolio project that demonstrates skills in AWS architecture, serverless development, conversational AI, NoSQL databases, analytics, and generative AI—highly relevant for Cloud Engineering and Solutions Architecture roles.